

Digital and Alternative Finance, FinTech Regulations in India

Unit 4

Financial Innovation

DEFINITION

- Financial innovation can be defined as an act of creating and then popularizing new financial instruments, technology , institutions , markets, process and business model including new application of existing idea's in different market context.

History of Financial Innovation

Key Historical Milestones in Financial Innovation

- **Ancient & Medieval Roots (3000 BCE – 1500s):** Recording agricultural loans on clay tablets in Mesopotamia, the Code of Hammurabi regulating interest (1800 BCE), and the introduction of metal coins in China (1000 BCE).
- **Birth of Modern Finance (1600s – 1800s):** The emergence of stock exchanges (Amsterdam, 1602) and corporate banking to fund maritime exploration and trade.
- **Technology & Computing Era (1950s – 1990s):**
 - **1950s-60s:** Introduction of credit cards and ATMs, reducing reliance on cash.
 - **1970s:** Launch of NASDAQ (first electronic stock market) and SWIFT (global messaging network).
 - **1980s:** Explosive growth in derivative products, options, and swap markets, fueled by computing power.
- **Digitalization & Fintech (2000s – Present):**
 - **1990s-2000s:** Rise of online banking and payment platforms like PayPal.
 - **2008-Present:** The 2008 financial crisis spurred development in FinTech, Blockchain and automated, low intervention investment tools, aiming for greater efficiency in borrowing and trading.

Drivers of Financial Innovation

Innovation is often propelled by the need to *manage risks* during volatile periods (like the 1970s), *regulatory changes*, and the demand for *increased speed*, as seen with modern flash trading. The shift has moved from *human-intensive banking to digitized, automated systems*.

Benefits of financial innovation

- Reduction in transaction cost.
- Moving funds across time and space-complete markets
- Managing risks and the pooling of funds.
- Boosting economic growth and development.
- Facilitating the sale and purchase of goods and services through a payment system.
- Addressing moral hazard and asymmetric information problem.

Causes for Financial innovation

- Low profitability
- Keen competition
- Economic liberalisation
- Improved communication technology
- Customer service
- Global impact
- Investors awareness

Types of Financial innovation

- Financial system/ institutional innovation
- Process innovation
- Product innovation

Motives of financial innovation

To provide ways of/ for:

1. Clearing and settling payments to facilitate trade.
2. The pooling of resources and for subdividing of shares in various enterprises.
3. Transfer economic resources through time , across borders and among industries.
4. Managing risk insurance.

Role of technology in financial innovation

Some types of financial innovation are driven by improvement in computer and telecommunications technology.

For most people, the creation of ATM was a greater financial innovation than assets backed securitization.

Others like debit and credit cards and online payment system.

These types of innovation are notable because they reduce transaction cost.

Examples of recent financial innovation

- Mobile payment system
- Peer to peer lending
- Online equity funding
- For worry free and hassle free banking
- For paying bills in a cinch
- E commerce

Objectives

Topic

Goal



Digitalization

- Learn about digitalization and digital goods
- Identify the characteristics of the digital customer
- Understand how products and services are digitalized



Digital Transformation in the Financial Sector

- Learn about new competitive dynamics in the digital era
- Realize the changes in the customer demand and expectations

Intro

- Financial services used to be one of the most conservative and stable industries
- Fintech industry flourishes: numerous fintech startups & internet giants direct some of their activities towards this direction
- The conversion of products of the industry to digital goods, the new customer demands, and the changing regulation.
- Financial industry is led to new era, the digital era of financial services and new rules

Digitalization

- Digital is becoming the way to do business
- Conversion of physical products and services into digital goods
- For the financial industry, digitalization implies that services offered do not involve any physical component:
 - E.g.,
 - cash is transformed into online and mobile payments
 - passbooks are transformed into apps

Digital goods

		Physical goods	Digital goods
Cost structure	■ Variable cost	High	Zero
	■ Fixed cost	It depends	High
	■ Distribution cost	High	Low
	■ Warehouse cost	High	Low
	■ Marketing cost	It depends	It depends
Pricing	■ Available pricing techniques	■ Limited pricing formulas ■ Usually constant values which depend variable costs	Several pricing approaches: ■ Based on usage ■ Fixed ("all you can eat") ■ Free (advertising) ■ Quantity discounts ■

The digital customer

<i>Characteristics</i>	<i>Description</i>
⌘ Connected	■ Is online ■ Use personal computers, tablets, smartphones, wearables to get informed and stay up-to-date 24/7
⌘ Informed	■ Access to a wide variety of information enables customers to compare prices, read product reviews and ratings, compare seller service ■ Put pressure on prices, quality of service and profit margins
⌘ Wary	■ Difficult to convince, skeptical on ads ■ Seek for other 'trusted' sources of information such as word-of-mouth, product reviews and personal recommendations
⌘ Vocal	■ Willing to voice their opinions, provide authentic expressions of their customer experience

History of the digitalization in the financial sector

- In the 60s, the financial sector began to use computers
- Late 60s, ATM machines
- Early 70s, worlds first electronic stock market
- Mid 70s, swift foundation established and provided secure messaging network
- Early 80s, first online bank (i.e., individuals and small businesses could check their account balance)
- Late 90s, rise of the Internet enables banks to provide online banking services through their website
- Late 90s, first mobile payments through SMS

The transformation of financial services

Transformation level	Banking IT (up to around 2008)	FinTech (after 2008)
External organization		
- Regulation	low equity requirements, low supervision	stricter rules; less protection
- Business model innovation	branch business & offline services	online & mobile services
- Governance of infrastructures	centralized institution as focal firm	distribution of tasks
- Payment style	majority of customers using cash	non-cash payments increase
Network organization		
- Networking	small number of network partners	many specialized partners
- Margins and cost structure	high margins in core business	lower margins, higher competition
- Competitors	other traditional financial service providers	start-ups, lateral entrants

The transformation of financial services

Transformation level	Banking IT (up to around 2008)	FinTech (after 2008)
- Culture	hierarchical	cooperative, agile
- Customer retention	high customer loyalty	reduced switching costs
Internal organization		
- Business focus	process-oriented	customer-centric
- Customer interaction	offline first	online first, omni-channel
- Core competencies	distribution, products, transactions	online distribution; platforms
- Vertical integration	high integration	low integration
- Service portfolio	banks are general service providers	small diverse providers
- Automation	processes require manual steps	fully-automated processes
- IT-architecture	monolithic systems, inhouse development	modular systems, APIs







What is Crowdfunding?

Crowdfunding is the practice of raising small amounts of capital from a large group of individuals, typically via online platforms, to fund a project, business, or charitable cause.

1. Donation-Based Crowdfunding (Charity)

- Definition:** Donors give money to a cause without expecting any financial or material return.
- Purpose:** Primarily for social causes, non-profits, disaster relief, or medical expenses.
- Examples:** GoFundMe, Ketto (India), Milaap.
- Key Aspect:** Driven by altruism and social impact.

2. Reward-Based Crowdfunding

- Definition:** Backers contribute funds in exchange for a non-monetary "reward" (perk), usually the product or service being developed.
- Purpose:** Used for creative projects, startups launching new products, or product pre-sales.
- Examples:** Kickstarter, Indiegogo.
- Benefits:** Market validation, community building, and zero equity dilution.

3. Equity-Based Crowdfunding (Investment)

- **Definition:** Investors receive small ownership shares (equity) in a company in exchange for capital.
- **Purpose:** High-growth startups or SMEs seeking larger capital (\$50k+).
- **Examples:** SeedInvest, CircleUp, LetsVenture (India).
- **Key Aspect:** Provides financial returns (dividends/appreciation) and requires regulatory oversight (e.g., SEBI in India).

4. P2P and Marketplace Lending (Debt-Based)

- **Definition:** Individuals lend money to borrowers (individuals or businesses) through online platforms, with the expectation of repayment plus interest.
 - **Marketplace vs. P2P:** Often used interchangeably, but "Marketplace" emphasizes the platform matching borrowers with investors, sometimes involving institutional investors, rather than just peer-to-peer. •
- Examples:** LendingClub, Prosper, Faircent (India), LenDenClub.
- **Benefits:** Lower costs than banks, quick, no ownership dilution.



Key Components for Success

- **Compelling Pitch:** A clear, concise narrative stating the problem and solution.
- **Video Content:** Videos significantly increase campaign performance. •
- **Transparent Goals:** Clearly stating how the funds will be used. •
- **Marketing & Communication:** Actively engaging the community.

Challenges & Risks

- **Fraud/Low Trust:** Risk of scams or underdelivering on promises.
- **Failure Rates:** Not all campaigns meet their funding goals.
- **Regulatory Hurdles:** Especially for equity and debt, legal compliance is necessary.

The Indian financial technology sector is experiencing a seismic shift in 2026, moving from a, "growth-at-all-costs" model to a phase defined by **governance, sustainability, and AI-native infrastructure**. While the early "Fintech" era focused on disrupting payments via mobile apps, the current "TechFin" phase involves large, tech-enabled entities leveraging massive datasets, regulatory compliance, and embedded finance to offer comprehensive, integrated services.

New Models: From Disruptors to Enablers

- **Invisible Finance:** Financial services are being woven into the fabric of non-financial platforms, making lending, insurance, and payments invisible at the point of intent.
- **AI-Native Operations:** Instead of "AI dashboards," firms are deploying agentic AI—systems that autonomously handle complex, multi-step workflows like underwriting, fraud detection, and regulatory reporting in real time.
- **Co-Lending & Secured Lending:** The lending model is shifting from unsecured, high-volume, quick disbursement loans to co-lending partnerships (bank/NBFC) that manage risk, with a stronger focus on secured loans for MSMEs.

- **Regulatory-First Approach (RegTech):** Compliance is no longer a back-office function but a competitive moat, with RegTech tools integrated directly into product workflows, reducing compliance costs by 30-50%.
- **Cross-Border Expansion:** The UPI infrastructure is going global (via projects like Nexus), transforming from a domestic payment rail into a global commerce bridge allowing instant, low-cost cross-border transfers.

New Products and Technologies

- **Credit Line on UPI (CLOU):** This allows users to access pre-approved credit lines directly within a "Scan & Pay" flow, making physical cards redundant.
- **Tokenized Asset Investments:** Leveraging blockchain to offer fractional ownership of real-world assets (RWA), such as real estate or private equity, making high-value investments accessible to retail users.
- **CBDC & Programmable Money:** The Reserve Bank of India (RBI) is scaling the e Rupee for government disbursements, B2B settlements, and cross-border remittances.
- **AI-Powered Fraud Defense:** Tools like **MuleHunter.AI** and behavioral analytics are being deployed to detect deepfakes, synthetic identities, and mule accounts in real time.
- **Embedded Insurance & Wealthtech:** Insurance products are integrated directly into marketplace checkouts (e-commerce/travel), while robo-advisors offer hyper-personalized portfolio rebalancing.

- **Account Aggregator (AA) Ecosystem:** With over 112 million users, AA acts as a standardized data-sharing layer, allowing non-bank fintechs to access high-quality data for underwriting

Key Drivers and Future Outlook

- **Profitability Focus:** With customer acquisition costs (CAC) tripled in some categories, the focus is on sustainable unit economics, with 2026 expected to be the year of IPOs for lending-tech firms.
- **Regional Expansion:** Over 35% of new fintech job postings are emerging from Tier 2 and Tier 3 cities, driven by the need for regional, language-specific UX and support.
- **Data Localization:** Stringent compliance requires that customer financial data must be stored within Indian borders, boosting local infrastructure providers.
- The 2026 landscape is, therefore, a shift toward "**fintech as infrastructure**," where the winners are those who can provide seamless, compliant, and trustworthy financial experiences rather than just fast, disruptive products.

The RBI is aggressively strengthening Fintech regulations in India (2025–2026) to enhance security, consumer protection, and orderly growth. Key measures include introducing a Self-Regulatory Organisation (SRO) framework, mandating Digital Lending App (DLA) registration by mid-2025, and proposing new rules to combat online fraud with compensation up to ₹50,000.

Key RBI Fintech Regulations & Amendments (2025-2026)

- **Self-Regulatory Organization (SRO) Framework (May 2024):** The RBI introduced a framework to foster collaboration, setting high standards for transparency, governance, and consumer protection in the fintech sector.
- **Digital Lending Guidelines (Updated 2025):** Stringent norms for Lending Service Providers (LSPs), mandating registration of all Digital Lending Apps with the RBI by June 15, 2025, and full compliance for multi-lender platforms by November 1, 2025.
- **Payment Aggregator (PA) Rules:** Non-bank entities must secure RBI authorization, with strict net worth requirements (increasing to ₹25 crore within 3 years).
- **Draft Customer Protection Rules (Effective July 1, 2026):** Proposes zero liability for customers in cases of bank negligence or prompt reporting, along with compensation for small-value frauds up to ₹50,000.
- **Fintech Repository & Cloud Facility:** A central repository is being developed to monitor tech stacks and activities, alongside an IFTAS-operated cloud facility to improve data security.

Key Policy & Structural Changes

- **UPI & E-mandates:** Increased UPI limits (up to ₹5 lakh) for specific categories and streamlined E-mandates for recurring payments.
- **Loan Regulations:** Draft directions on co-lending, gold-backed loans, and non-fund-based credit to enhance transparency and risk management.

- **NBFC Compliance:** Enhanced Net Owned Funds (NOF) requirement to ₹10 crore for NBFCs by March 31, 2027.

These measures indicate a transition from a light-touch approach to a robust, compliance-driven framework for Indian fin techs.

RESTRICTIONS

The RBI issued the Master Direction on Outsourcing of Information Technology Services in April 2023, which akin to the extant regulatory frameworks on outsourcing, reaffirm that the outsourcing of arrangements does not undermine the responsibility of the reporting and regulated entities. While these directions permit outsourcing of non-core functions, and prohibit the outsourcing of core and critical functions, they mandate that the responsibility *vis-à-vis* the customers continue to vest with the regulated entities. These directions impose requirements pertaining to establishing a grievance redressal framework, obligations of due diligence, monitoring and risk management. The RBI has tightened its clutches on outsourcing arrangements, and the requirements of having a robust business continuity plan and enabling audits by the RBI and the regulated entities for such service providers have been the most critical discussion points.

While the RBI expressly recognised the outsourcing of lenders' functions to Lending SPs under the Digital-Lending Guidelines issued on September 2, 2022, the RBI has prescribed a host of restrictions on Lending SPs. These primarily relate to Lending SPs' use and access of customers' data. The Lending SPs cannot store personal information of customers, except some basic minimal data (*viz.*, name, address, contact details of the customer, etc.) required to carry out the lender's operations. A specific prohibition on storage of biometric data, often used for KYC measures, has also been envisaged.

In June 2023, the RBI issued guidelines to regulate the market practice of first loss default guarantees (“**Default Guarantee**”) provided by Lending SPs or other regulated entities. Default Guarantee refers to contractual arrangements where the Default Guarantee provider guarantees to compensate a regulated entity for a percentage of losses occurring due to loan defaults.

While until now the regulation of Default Guarantee for digital lending arrangements was unclear, the new guidelines allow such arrangements but impose certain requirements regarding permissible structures for Default Guarantee, caps on coverage, invocation

procedures, and due diligence and disclosure obligations.

CROSS-BORDER BUSINESS

With the objective of making cross-border payments seamless and efficient, like the domestic payments, the RBI and NPCI has given special importance to FinTech offerings that have witnessed immense success domestically, such as the offerings that have UPI, RuPay card network and Bill Payment-System as the backbone. The year 2023 was one of regulatory innovation and introduction of regulations for new-age business models particularly in cross-border remittances space, including through UPI or Aggregators.

The journey of UPI from its inception to its global expansion is a testament to India's digital innovation. Within India, UPI's growth has been exponential, with transactions soaring from 920 million in FY 2017–18 to 83.8 billion in FY 2022–23, and transaction values rising from USD 12 billion to USD 1.67 trillion in the same period.

However, besides the prominence in the domestic market, the UPI ecosystem has made various cross-border strides in the recent years, marking a significant milestone, showcasing India's technological advancements and setting the stage for a more interconnected global payment system. Some of these key measures include:

1. One of the most crucial regulatory developments in the cross-border space by the RBI was the step of bringing the Cross-Border Aggregator, *viz.*, the entities that facilitate cross-border payments for import and export of goods and services in online mode, under its direct regulation. The RBI prescribed an authorization requirement for the Cross-Border-Aggregators and laid out certain operational conditions for facilitation of payments for cross-border transactions.
2. In June and July 2024, NPCI's International Payments arm has engaged in partnership with Network International, a leading enabler of digital commerce across the Middle East and Africa (MEA region). This partnership will enable acceptance of QR code-based UPI

payments through Network's POS terminals in the UAE.

3. Continuing the expansion of the UPI-like model to other countries to enable deployment of a UPI-like real-time payments system, NPCI has partnered with the Bank of Namibia in May 2024 for Namibia and the Central Reserve Bank of Peru in June 2024 (with Peru becoming the first country in South America to do so).
4. The RBI has collaborated with the Bank for International Settlements and central banks of four ASEAN countries (Association of Southeast Asian Nations) for a multilateral initiative to enable retail cross-border payments. This will interlink domestic payment systems of India with those of Malaysia, Philippines, Singapore and Thailand.
5. The Government of India has executed agreements and memorandums with various countries (like Russia) to enable acceptance of electronic payments through RuPay cards, a proprietary card network developed by NPCI, in other countries.
6. The payment system for utility bill payments operated by the NPCI, *viz.*, Bill Payment-System has also been opened for international payments, enabling a person residing in select identified countries to make bill payments for utility bills in India.

Fintech companies face significant regulatory challenges in 2026, primarily driven by navigating fragmented, rapidly evolving global regulations, stringent data privacy laws, and intense cybersecurity threats. Key hurdles include obtaining diverse operational licenses, ensuring robust Anti-Money Laundering (AML/KYC) compliance, and balancing rapid innovation with strict oversight.

Key Regulatory Challenges for Fintech

- **Evolving & Fragmented Regulations:** FinTech must adapt to constantly changing guidelines that often lag behind technological advancements, creating a "catch-up" scenario. The lack of uniform, global regulations leads to complex compliance, especially for cross-border operations.

- **Data Privacy & Cybersecurity:** Handling vast amounts of sensitive customer information makes fin techs prime targets for cyberattacks, requiring stringent, evolving data governance.
- **Licensing & Compliance Costs:** Securing appropriate licenses (e.g., banking, payment institution) is expensive and time consuming. Compliance costs are high, particularly for startups with limited resources.
 - **Anti-Money Laundering (AML) & KYC:** Stricter enforcement of AML and Know Your Customer (KYC) regulations requires rigorous, real-time monitoring.
- **Digital Assets & AI Regulation:** Emerging technologies like decentralized finance (DeFi), blockchain, and AI pose unique challenges, with regulators increasingly focusing on stablecoin issuers and algorithmic accountability.

Strategies for Navigating Compliance

To mitigate these risks, companies are investing in RegTech solutions, Automating, establishing dedicated compliance teams, conducting regular audits, and leveraging regulatory sandboxes.